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Spontaneous switching in a protein signalling array reveals near-critical cooperativity

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Johannes M. Keegstra [1,2,4], Fotios Avgidis [1,4], Evan Usher [1], Yuval Mulla [1], John S. Parkinson [3] & Thomas S. Shimizu [1]

Cooperative interactions within large protein assemblies are crucial for cellular information processing. However, direct observations of cooperative transitions have been limited to compact molecular assemblies. Here we report the in vivo measurements of spontaneous discrete-level transitions in the activity of an entire *Escherichia coli* chemosensory array—an extensive membrane-associated assembly comprising thousands of molecules. Finite-size scaling analysis of the temporal statistics reveals nearest-neighbour coupling strengths within 3% of the Ising phase transition, indicating that chemosensory arrays are poised at criticality. We also show how *E. coli* exploits both static and dynamic disorder, arising from chemoreceptor mixing and sensory adaptation, respectively, to temper the near-critical dynamics. This tempering eliminates detrimental slowing of response while retaining substantial signal gain as well as an ability to modulate physiologically relevant signal noise. These results identify near-critical cooperativity as a design principle for balancing the inherent trade-off between response amplitude and response speed in higher-order signalling assemblies.

Large protein assemblies at the heart of many cell signalling processes exhibit varying degrees of structural and dynamical order, from liquid-like granules[1–3] to solid-like arrays[4–6]. Recent experiments are revealing how phase transitions that lead to the formation of liquid-like assemblies—discontinuous phase transitions analogous to the condensation of gas into liquid—are used by cells to implement various information processing tasks, such as signal initiation and confinement[7], kinetic proofreading[8] and noise control[9]. Theory also predicts that different phase transitions can occur within solid-like assemblies—continuous phase transitions, analogous to the ordering of magnetic spins in a ferromagnet at low temperatures—arising from conformational interactions between protein subunits[10]. However, requirements for continuous phase transitions are more stringent than those for discontinuous phase transitions—they occur only at a special point in phase space—a critical point. It remains unclear if such critical transitions occur in signalling assemblies and, if so, how they constrain or contribute to the functional design of protein assemblies. Elucidating the physical principles of signalling assemblies would not only advance our understanding of cell signalling in nature but also provide new design principles for the rational design of synthetic protein circuits[11–14].

Recent structural studies are revealing an increasing number of solid-like signalling assemblies with a high degree of spatial order, where subunit proteins are arrayed in a regular pattern. The repertoire of such crystal-like assemblies reported so far is diverse in both form and function, and includes one-dimensional filament-like assemblies found in cellular homeostasis and inflammation signalling[15,16], protein rings that mediate the control of cell motility and apoptosis[17,18], as well as two-dimensional arrays involved in chemosensing, neuromuscular control and innate immune responses[19–22]. Yet, despite exciting advances in resolving the ultrastructure of these large assemblies[23,24], the mechanistic design principles of their signalling function remain challenging to address experimentally. Of particular interest is the ability of signalling assemblies to perform signal processing by way of cooperative interactions between assembly subunits. In contrast to the compact oligomeric signalling complexes of fixed size, these extensive structures tend to assemble through open-ended polymerization and/or

[1]AMOLF, Amsterdam, The Netherlands. [2]Institute for Environmental Engineering, ETH Zurich, Zurich, Switzerland. [3]School of Biological Sciences, University of Utah, Salt Lake City, UT, USA. [4]These authors contributed equally: Johannes M. Keegstra, Fotios Avgidis. e-mail: shimizu@amolf.nl

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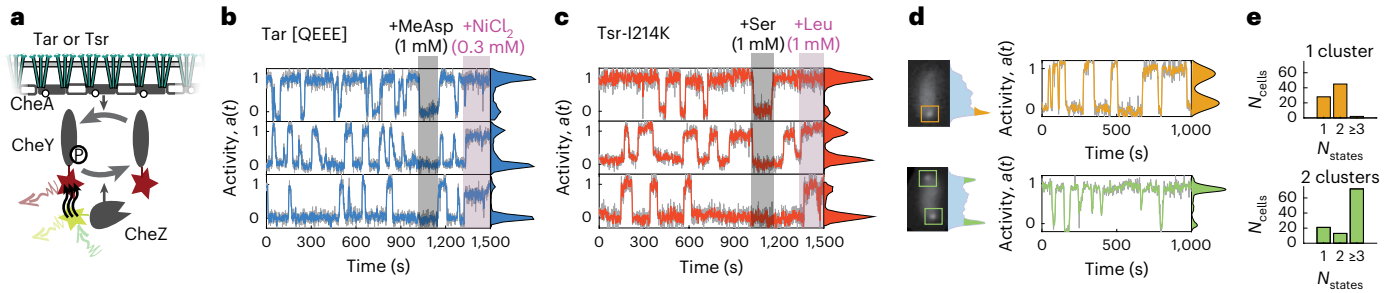

Fig. 1 | *In singulo* measurements of chemosensory array dynamics reveal two-state switching fluctuations. a, Illustration of single-cell FRET assay to measure the kinase activity[34], with a schematic of the side view of a membrane-associated chemosensory array of chemoreceptors (either Tar or Tsr). The kinase (CheA) within the chemosensory array phosphorylates CheY (CheY-P), whereas CheZ dephosphorylates CheY. The FRET signal (black arrows) between fluorescently labelled CheZ and CheY, measured from the emitted fluorescence intensities (red and yellow arrows) is proportional to the activity of the chemosensory array. **b**,**c**, Activity time series from single-cell FRET experiments that reveal two-state switching in the absence of any chemoeffector, of representative cells without adaptation enzymes expressing only the chemoreceptor Tar [QEEE] (**b**; blue) or the chemoreceptor Tsr-I214K (**c**; red). Cells are exposed to the measurement buffer (MotM) for most of the experiment. To obtain the minimum and maximum FRET levels, cells are exposed to an attractant stimulus (Tar: 1 mM α-methylaspartate (MeAsp); Tsr: 1 mM L-serine (Ser), grey bands) and a repellent stimulus (Tar: 0.3 mM $NiCl_2$; Tsr: 1 mM L-leucine (Leu), purple bands). These levels were then used to normalize the activity time series of each cell between zero and unity. Example time series other than the two-state one are shown in Extended Data Fig. 1. **d**, Chemosensory cluster organization from CheZ localization (left) and the corresponding activity time series (right, raw in grey, 6 s (averaged); in colour) and histograms (on the right margin) of activity determined by FRET, for a representative cell expressing only the chemoreceptor Tsr-I214K, with a single visible cluster exhibiting two-state switching (orange) and a representative cell with two visible clusters exhibiting multistate switching (green). Additional time series and cluster organization are shown in Extended Data Fig. 2. **e**, Histograms of the number of states in switching cells, for cells with one visible cluster (top, orange) and for cells with two visible clusters (bottom, green).

multivalent interactions, making them inherently variable in both size and composition[5]. The resulting polydispersity and stoichiometric diversity tend to mask their true signalling dynamics[25,26], as well as their size and composition dependencies[6,27], both in vivo and in vitro. An ideal functional assay would, therefore, interrogate assembly level dynamics *in singulo*—at the level of an individual assembly—but experimental realization has remained elusive.

Here we report in vivo FRET experiments that resolve this challenge for the chemosensory array of *Escherichia coli*, a canonical two-dimensional signalling assembly lining the cytoplasmic membrane, which allows motile bacteria to bias their random-walk swimming trajectories to navigate chemical environments[28–30]. This higher-order assembly comprises thousands of receptor, kinase and scaffolding molecules arranged in a well-defined lattice structure[31–33] and, thus, serves as a paradigm for signalling in two-dimensional protein assemblies. The array integrates input signals (chemoeffector ligand concentrations) with a negative feedback signal (covalent modification state of receptors) to generate a signal output (activity of the kinase, CheA) that can be followed in real time by an intermolecular Förster resonance energy transfer (FRET) system[28]. By labelling two downstream proteins, the response regulator CheY, which is phosphorylated by CheA, and its phosphatase CheZ, which dephosphorylates CheY, a FRET read-out proportional to the kinase activity can be obtained within live cells[28] (Fig. 1a). Our strategy leverages recent advances in extending this in vivo FRET technique to the single-cell level[34,35], to achieve functional measurements of individual arrays within live cells. Although this FRET system provides a whole-cell measurement that integrates the kinase output of all arrays within each cell, the number of arrays per cell is highly variable[36] and in a substantial minority of cells, nearly all receptors and other array-component proteins are assembled within one dominant large array[37]. Thus, by searching for cells in which signalling is dominated by a single large array, we aimed to achieve *in singulo* FRET measurements of functional array output.

Spontaneous two-state switching in chemosensory arrays

To identify cells whose chemoreceptor population is concentrated into a single large array, we focused on fluctuations in the kinase output that are observable by FRET at the single-cell level. In wild-type (WT) *Escherichia coli* cells, the stochastic kinetics of the reversible covalent modification reactions (methylation/demethylation) mediated by the adaptation system are known to drive temporal fluctuations in kinase activity under constant environmental conditions[34,35,38]. Surprisingly, however, previous single-cell FRET studies identified the largest fluctuations in engineered genetic mutants devoid of the adaptation system and expressing only one of the five chemoreceptor species[34,35], with a subpopulation of cells exhibiting two-level (all-ON/all-OFF) fluctuations in kinase output as the input ligand signal was held constant, at a sub-saturating level[34] (Supplementary Note 1). If these fluctuations were driven by the intrinsic dynamics of the array, such synchronized two-level stochastic switching of the entire kinase population would be expected only if whole-cell signalling were dominated by a single dominant array. However, because an external ligand was present in those experiments, the possibility remained that the observed output switches were caused by spurious fluctuations in the external chemoeffector signal. We therefore began by testing whether two-level fluctuations occur in the absence of any external ligand stimulus by exploiting known genetic modifications of chemoreceptors that mimic the effects of methylation/demethylation, and can modify the chemoreceptor activity bias and maintain otherwise normal signalling function[39]. We tuned down the activity bias by expressing the aspartate receptor Tar in the QEEE covalent modification state (from QEQE in WT Tar), which yields an intermediate activity bias without the addition of chemoeffectors (Supplementary Note 1 and Extended Data Fig. 1e)[39]. This allowed us to measure temporal fluctuations by single-cell FRET recordings in thousands of individual cells (Fig. 1b and Extended Data Fig. 1a). In 204 out of 1,414 cells obtained from 19 FRET experiments, we observed two-level switching between a high- and low-activity state (>65% of transitions showing activity-level changes of >0.7). Spontaneous switching in the absence of exogenous ligands was not specific to Tar receptors, as we observed a similar switching behaviour (two-state switching in 548 out of 4,446 cells across 44 experiments) in analogous experiments with Tsr-I214K, a single-residue replacement mutant of the serine receptor Tsr within its 'control cable' region[40] with a downshifted activity bias similar to that of Tar [QEEE] (Fig. 1c and Extended Data Fig. 1b).

Spontaneous two-state switching is a hallmark of allosteric signalling complexes such as ion channels[41], but their observation requires

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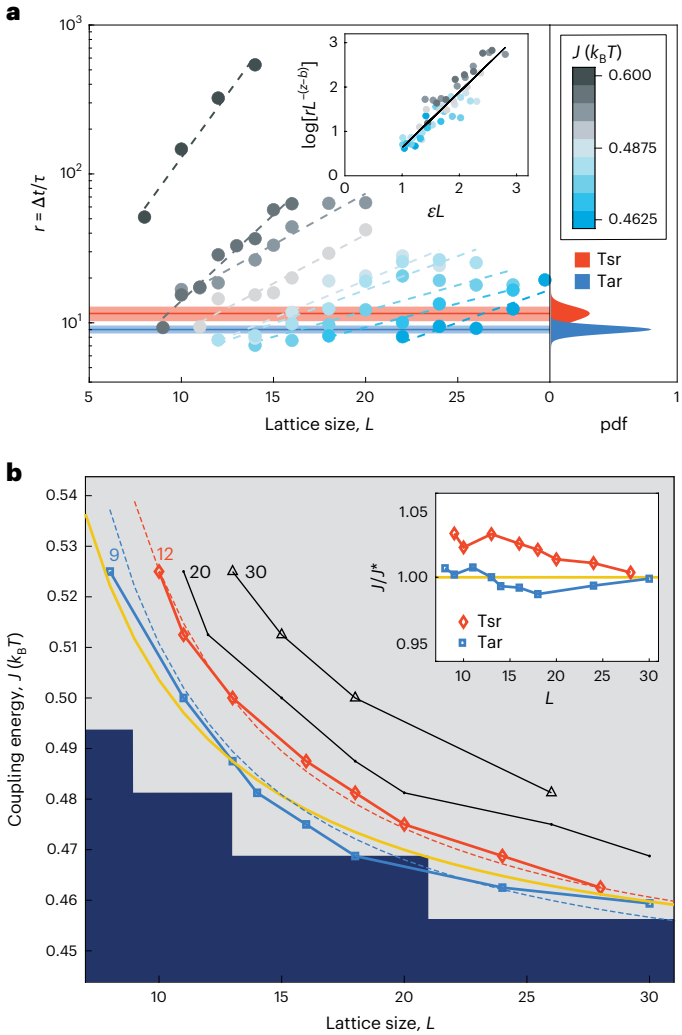

Fig. 3 | Finite-size scaling analysis reveals near-critical cooperativity of chemosensory arrays. a, Left: switching timescale ratio $r = \langle \Delta t \rangle / \langle \tau \rangle$ for various values of the coupling energy J (blue to black) as a function of lattice size L (circles), with exponential fits (dashed lines). Inset: data collapse for the near-critical region ($J < 0.6 k_B T$), on rescaling by factors $\epsilon = |J^* - J|/J$ and $L^{-(z-b)}$ according to the finite-size scaling theory (Extended Data Fig. 7 shows the determination of scaling parameters). Right margin: estimated distribution (Supplementary Fig. 8) of the switching timescale ratio $\langle \Delta t \rangle / \langle \tau \rangle$ across cells, for Tsr-I214K (red) or Tar [QEEE] (blue). The shaded areas (left margin) correspond to one standard deviation of the estimated timescale ratio distribution. **b**, Phase diagram of the switching behaviour as in the L–J plane. Parameter value pairs generated polarized (grey region) and non-polarized (dark blue region) fluctuations as estimated from the peak-valley ratio of the time series histograms. Isolines of similar switching timescale ratios, as indicated, were constructed from the simulations (beaded coloured and black curves). Theoretical finite-size scaling predictions for the critical coupling energy $J^*(L)$ (ref. 62; gold curve) and the switching time ratios (dashed lines; Supplementary Equation (9) and Supplementary Table 16 list the fit parameters) are superimposed. Inset: switching timescale ratio isolines for Tar (blue) and Tsr (red), normalized by the finite-size scaling of the critical coupling energy J^* reveals values within 3% of the critical energy.

isolines $J_r^{\mathrm{iso}}(L)$ corresponding to the measured r values for Tar ($r_{\mathrm{Tar}} \approx 9$; Fig. 3b, blue) and Tsr ($r_{\mathrm{Tsr}} \approx 12$; Fig. 3b, red) revealed a remarkable confinement of the coupling energy for both Tar and Tsr to within ±3% of $J^*(L)$ across a broad range in L (Fig. 3b, inset). Thus, despite uncertainty in the array size L, finite-size scaling analysis of the observed temporal statistics strongly suggests that both Tar and Tsr arrays are poised very close to the Ising critical point.

Arrays without adaptation feedback exhibit critical slowing of response

Finite-size scaling analysis also allowed us to estimate the fundamental flipping timescale $1/\omega_0$ of allosteric units. Combining the observed residence and transition times with an approximate range for the size L based on the known structure, stoichiometry and component expression levels of the array, the scaling could be calibrated to yield $1/\omega_0 \approx 10$–35 ms (Extended Data Fig. 8 and Supplementary Note 3). This allowed us to directly compare the response dynamics between simulations and experiments. Previous theoretical studies have shown that signal amplification due to coupling within Ising-type conformational spread models comes at the cost of prolonged response times[47,63], potentially impairing temporal comparisons during run-and-tumble navigation[64–66]. To address the influence of near-critical cooperativity on response times to external stimuli, we first simulated the dynamic response of arrays with $J \approx J^*$ after an applied step stimulus (implemented within the model by an external ligand field ΔH that modifies the flipping rate ω_0 by a factor of $e^{\Delta H(2a-1)/2}$; Methods) to mimic kinase inactivation on attractant chemoeffector stimulation. These simulations demonstrated that near-critical Ising arrays indeed respond slowly, with a broad distribution of times averaging tens of seconds (Fig. 4a). We then conducted FRET experiments with a sub-saturating stimulation of adaptation-deficient cells expressing homogeneous arrays (consisting of a single chemoreceptor species, Tsr) using a polydimethylsiloxane (PDMS) microfluidic flow cell that allowed for chemoeffector concentration changes much faster than the acquisition frequency of 1 Hz (Methods). The experimental response time courses (Fig. 4b) were remarkably similar to those of the array simulations (Fig. 4a), thereby confirming that the drastic slowing of response can indeed occur in *E. coli* homogeneous chemosensory arrays near criticality.

Chemosensory arrays in WT *E coli* cells are known to mix multiple chemoreceptor species[67], and at the population level, such receptor mixing has been shown to attenuate (though not eliminate) ligand–response cooperativity[48]. Considering our current finding that the strength of nearest-neighbour interactions within homogeneous arrays are near critical ($J \approx J^*$) for both Tar and Tsr arrays, attenuated cooperativity upon mixing Tar and Tsr suggests that J might take on a different value between lattice sites occupied by homologous and heterologous receptors (that is, $J_{\mathrm{Tar-Tar}} \approx J_{\mathrm{Tsr-Tsr}} \approx J^*$, but $J_{\mathrm{Tar-Tsr}} \neq J^*$). Thus, receptor-species mixing introduces a spatial disorder in coupling (that is, a 'bond disorder') that potentially limits the extent of cooperative interactions and, therefore, attenuates critical slowing down (Supplementary Note 4). Experimentally, the degree of chemoreceptor mixing can be varied by controlling cell culturing conditions. Cells with a WT chemoreceptor complement are known to have a relatively low degree of mixing when harvested at an early exponential phase (array dominated by Tsr), but a higher degree of mixing when harvested at mid-exponential phase (increased Tar)[68,69] (Supplementary Note 5). FRET experiments on adaptation-deficient cells harvested at these different growth phases revealed that although receptor mixing can indeed attenuate response slowdown (Fig. 4c,d,e), responses to sub-saturating stimuli at early exponential phase can still be very slow (>10 s), limiting the speed at which bacteria can respond. Taken together, these results provide a clear experimental demonstration that cooperative interactions within the array affect the signalling response timescale of *E. coli*. Although cooperativity-induced response slowdown is most acute in cells engineered to express only a single chemoreceptor species, it is substantial also in cells with the WT complement of chemoreceptor species.

Near-critical cooperativity balances a speed–amplitude trade-off

Why are bacterial chemosensory arrays poised so close to criticality, despite potentially deleterious slowing of response? To address this question, we examined the relationship between response speed and

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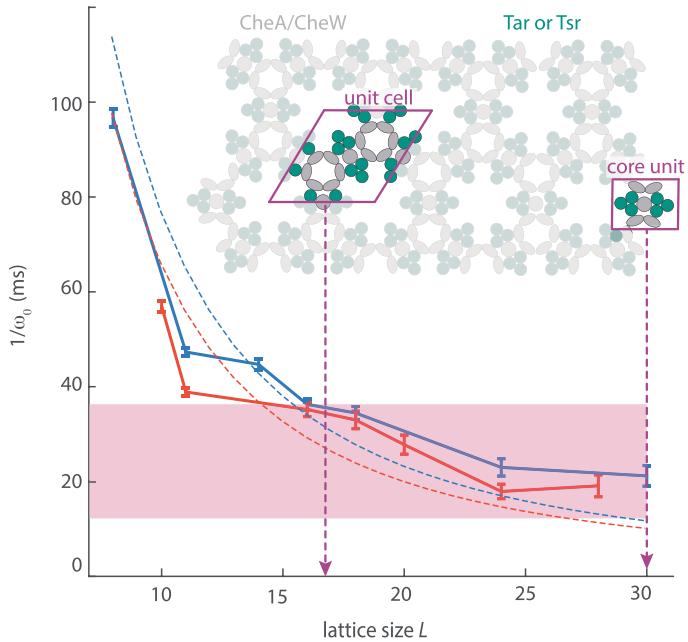

Extended Data Fig. 8 | Estimation of the intrinsic fluctuation timescale of allosteric units. Fundamental transition timescale $1/\omega_0$ of allosteric units, plotted as a function of the assumed array size L for Tar (blue) or Tsr (red) arrays, based on simulations calibrated by experimentally observed residence- and transition-times (solid lines). Shown together are the expected dependence based on finite-sized scaling theory (blue and red dashed lines; see Supplementary Note 2.6). Two possible fundamental units of cooperativity in the bacterial chemosensory array are highlighted within a schematic top view of chemosensory arrays: 'core unit' consisting of a CheA kinase dimer, two CheW monomers, and two trimers of chemoreceptor dimers (purple rectangle) and the repeating unit cell of the array, consisting of one full core unit and four half core units (purple parallelogram). Each chemosensory array contains approximately 1000 core units or 300 repeating unit cells, leading to lower and upper limits of array size (purple dashed lines). Using these as lower and upper limits yields a timescale range $1/\omega_0 \approx 15-35$ ms (shaded area). Error bars refer to s.e.m.